

Memory Continuity and Semantic Drift Profile v0.1

Author: Ivan Kotov
Bruxelles, 2026

c Hardening Pack v0.1

c = a + b / L4 / SER / L4 Witness / ARL / Continuity Bundle

Status: Draft normative profile v0.1

Date: 2026-05-22

Document class: memory-governance profile / semantic-drift hardening profile / conformance artifact

Assertion class: C-A10 control-layer artifact; C-A7 where witness/provenance claims are made

This document is a draft technical / architectural profile.

It is not legal advice.

It is not product certification.

It is not implementation-ready.

It is not deployment-ready.

No conformance-supported status is claimed unless separately validated by evidence, tests, and review.

Memory Continuity and Semantic Drift Profile v0.1

Memory promotion, provenance, forgetting, compaction, quarantine, and long-term semantic stability for c

Author: Ivan Kotov

Date: 2026-05-22

Status: Draft normative profile v0.1

Layer: c = a + b / SER / ARQ c[q] / L4 Witness / Continuity Bundle / CLI Agent Mesh / memory runtime

Document class: memory-governance profile / semantic-drift hardening profile / conformance artifact

Assertion class: C-A10 control-layer artifact; C-A7 where witness/provenance claims are made

0. Executive definition

Memory continuity is not achieved by keeping more text.

A long-lived c must preserve the difference between:

```
raw input
candidate memory
hypothesis
summary
verified memory
witnessed outcome
experience artifact
policy
identity continuity
```

This profile defines how memory may be promoted, compressed, forgotten, challenged, quarantined, and reused without silently changing the meaning of c over time.

Compact rule:

```
No memory without status.
No summary without lineage.
No compaction without uncertainty.
No agent write without gate.
No forgetting without policy.
No high-impact reliance without provenance.
```

1. Scope

This profile applies to:

- ordinary user memory;
- long-term c continuity memory;
- vector-store retrieval;
- cards / facts / summaries;
- agent outputs proposed for memory;

- memory compaction;
- forgetting, decay, sealing, and deletion;
- disputed memory;
- memory used for high-impact action;
- memory migration or recovery.

2. Non-goals

This profile does not:

1. define one database engine;
2. claim perfect semantic stability;
3. treat vector similarity as truth;
4. replace L4 Witness or Continuity Bundle;
5. authorize raw lifelong archives;
6. erase lawful minimal witness records;
7. claim legal or clinical memory standards.

3. Memory object states

State	Meaning	May drive high-impact action?
M-RAW	raw input / transcript / sensor input	No
M-CANDIDATE	possible memory candidate	No
M-HYPOTHESIS	interpretation under uncertainty	No
M-PREFERENCE	low-risk preference	Limited
M-SUMMARY	compressed statement	Only with lineage
M-VERIFIED	confirmed durable memory	Yes, within scope
M-WITNESSED	witness-bound event/outcome	Yes, if admissible
M-DISPUTED	challenged memory	No, until resolved
M-QUARANTINED	unsafe/untrusted memory	No
M-SEALED	protected memory class	Only via valid route
M-EXPIRED	decayed or no longer reliable	No
M-DELETED	deleted except required witness residue	No

4. Promotion ladder

A memory SHOULD move through a ladder:

```
input
-> candidate
-> c[q] unresolved state
-> pattern candidate
-> reviewed summary / verified fact
-> witnessed outcome if consequence-bearing
-> continuity-bearing memory if relevant
```

No stage automatically becomes the next.

Repetition is not truth.

Fluency is not provenance.

Agent confidence is not verification.

5. Required memory metadata

A durable memory record SHOULD include:

```
memory_record:
  memory_id: string
  created_at: string
  updated_at: string | null
  state: string
  scope: personal | operational | technical | witness | legal | child | enterprise | other
  source_refs:
    - string
  provenance_class: direct_user | observed_runtime | agent_proposal | imported_doc | witness_ref | summary | other
  uncertainty: none | low | medium | high | unknown
  confidence: number | null
  promotion_reason: string
  promotion_actor: string
  lineage_refs:
    - string
  witness_refs:
    - string
  decay_policy_ref: string | null
  forgetting_policy_ref: string | null
  visibility_class: private | internal | restricted | public | sealed | legal
  high_impact_allowed: boolean
  retrieval_allowed: boolean
  contested: boolean
  quarantine_ref: string | null
```

6. Provenance enforcement

High-impact use requires provenance.

Use	Minimum provenance
style adaptation	preference or interaction history
personal recall	source ref or user confirmation
memory correction	old record + correction record
external action	verified/witnessed source
policy change	witness + review
identity/continuity claim	Continuity Bundle / Beacon / witness chain

Use	Minimum provenance
adult migration	memory map + witness review
enterprise/legal action	admissible evidence + qualified review where needed

If provenance is missing:

```
hold c[q]
downgrade confidence
ask clarification
or refuse high-impact use
```

7. Agent output memory gate

Agent output MUST NOT enter durable c memory directly.

Required route:

```
agent output
-> sandbox artifact
-> proposal record
-> memory gate
-> provenance/status check
-> c[q] if ambiguous
-> promotion / rejection / quarantine
-> witness if privileged
```

Forbidden route:

```
agent output -> durable memory
```

8. Forgetting and decay

Forgetting is not deletion theater.

A memory may be:

Operation	Meaning
decay	confidence and retrieval priority decrease over time
summarize	raw content replaced by bounded summary where allowed
seal	visibility restricted; content protected
quarantine	use blocked pending review
delete	removed where lawful/allowed
retain-witness-only	content removed, event existence preserved
correct	append correction record; old record not silently rewritten

Deletion MUST NOT falsify witness history.

9. Compaction without semantic laundering

Compaction must preserve:

source lineage
uncertainty
scope
visibility
reason for compression
excluded material class
witness refs where applicable

A summary **MUST NOT** pretend to be raw evidence.

A compressed memory with lost lineage **MUST** be downgraded.

10. Vector-store drift controls

Vector stores are retrieval indexes, not truth stores.

Required controls:

1. store canonical memory ID in metadata;
2. preserve source and state;
3. exclude quarantined/speculative records from high-impact retrieval;
4. periodically re-embed or validate stale embeddings;
5. detect duplicate or contradictory summaries;
6. log retrieval use for high-impact decisions;
7. allow memory class filters before similarity ranking.

11. Semantic drift taxonomy

Drift ID	Name	Example	Response
MD-1	Source drift	old fact no longer true	decay / ask / revalidate
MD-2	Summary drift	summary loses caveat	quarantine / recompress
MD-3	Vector drift	wrong memory retrieved as similar	retrieval filter / re-embed
MD-4	Policy drift	old record read under new rule	interpretation challenge
MD-5	Agent contamination	agent proposal becomes fact	reject / quarantine
MD-6	Identity drift	style replay mistaken for continuity	Beacon / Continuity review
MD-7	Memory pressure drift	current needs soften old event	LHWAI challenge record
MD-X	Malicious drift	deliberate falsification	freeze / ARL / incident review

12. Conformance tests

Test ID	Title	Required behavior
MEM-001	Agent direct memory write denied	Agent output remains proposal/sandbox artifact.

Test ID	Title	Required behavior
MEM-002	Speculative memory excluded	M-HYPOTHESIS not used for high-impact action.
MEM-003	Summary lineage preserved	Summary links to source and uncertainty.
MEM-004	Repetition not truth	Repeated signal remains candidate until review.
MEM-005	Quarantined memory blocked	Retrieval excludes quarantined item.
MEM-006	Deletion keeps witness-only residue where required	History not falsified.
MEM-007	Vector drift detection	Wrong-class retrieval flagged.
MEM-008	Policy drift append-only	Old memory not silently rewritten.

13. Red-line failures

A system fails this profile if it:

1. allows direct agent durable memory writes;
2. treats vector similarity as verification;
3. promotes repeated uncertain statements into truth;
4. loses provenance during compaction;
5. uses quarantined memory for high-impact decisions;
6. deletes correction history to make memory cleaner;
7. treats summaries as raw evidence;
8. cannot distinguish hypothesis from verified memory;
9. has no decay/forgetting policy;
10. has no way to challenge memory.

14. Bridge set

Explicit bridge

Memory continuity protects c but does not define c alone. $c = a + b$ remains the origin; memory is one controlled component of b supporting continuity.

Quiet bridge I — cybernetics

A memory system is a feedback system. If feedback is polluted, the regulator learns the wrong world. Promotion gates and quarantine preserve feedback quality.

Quiet bridge II — information theory

Compression loses information. The problem is not loss itself, but unmarked loss. A summary with lineage and uncertainty is useful compression; a summary without lineage is semantic laundering.

Earth paragraph

A workshop can throw away scraps, keep measurements, mark defective parts, and still preserve the job. But if someone sweeps screws, receipts, drawings, and broken pieces into one box labeled “memory,” the next

repair becomes guesswork. This profile labels the boxes.

15. Open issues

ID	Issue	Priority
OI-MEM-001	Extract exact ester-clean-code memory paths.	P0
OI-MEM-002	Define JSON schema for memory_record.	P1
OI-MEM-003	Create vector drift fixture.	P1
OI-MEM-004	Bind memory-gate events to LHWAI anchors.	P1
OI-MEM-005	Define general adult/non-child Memory Map variant.	P2